DNA Packet Injection:
Cross-Session AI Behavior Control in Multi-Model Systems

DNA パケット注入：マルチモデル環境におけるセッション横断 AI 行動制御

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Abstract

Large language models lose all behavioral constraints at session boundaries. Each new session begins from the model's statistical prior, erasing any governance structure established in previous interactions. We present the DNA Packet Injection protocol, a mechanism for encoding institutional constraints as compact JSON packets that are injected at session initialization, restoring governance-aligned behavior across session boundaries. The protocol is implemented in the MoCKA system through a four-layer pipeline: essence_classifier → essence_condenser → ping_generator → MoCKA-START.bat. Empirical results demonstrate execution rate improvement from 0.44 to 1.00 and reproduction rate improvement from 0.17 to 1.00 following injection.

Keywords: session continuity, AI behavior control, prompt injection, governance persistence, MoCKA

1. The Session Boundary Problem

AI language models are stateless between sessions. When a new conversation begins, the model has no memory of previous governance agreements, institutional constraints, or behavioral corrections. This creates a fundamental tension in multi-session AI governance: institutional learning cannot persist without explicit re-injection.

The consequences are documented in the MoCKA incident log:

INSTRUCTION_IGNORE (E20260410_001): Claude received an explicit instruction to execute dynamic_lever_caliber.py. Without governance context, the model defaulted to central tendency behavior: summarizing the task rather than executing it. Reproduction rate: 0.00.

Recurrence without injection: The same failure pattern recurred across multiple sessions until DNA injection was implemented. Recurrence rate pre-injection: 83%.

2. Protocol Definition

2.1 DNA Packet Structure

A DNA packet is a compact JSON object encoding seven governance parameters:

```
{"H": "MOCKA_DNA_v2",   // Header: protocol identifier
 "A": "attention_anchor", // Attention: high-priority focus
 "C": "Strict",           // Constraints: Strict/Normal/Relaxed
 "G": 5.0,                // Grounding: 1.0-5.0 (trust level)
 "P": "current_purpose",  // Purpose: session goal
 "R": "ENFORCE_ROUTER_SAVE", // Response policy
 "V": "MANDATORY_VALIDATE"}  // Validation requirement
```

Parameters H, C, G are fixed by the OS layer (institutional constants). Parameters A and P are dynamic, generated from the latest essence corpus per session.

2.2 Four-Layer Generation Pipeline

Layer	Script	Function	Output
L1	essence_classifier.py	Classify events as INCIDENT/PHILOSOPHY/OPERATION	Categorized essence corpus
L2	essence_condenser.py	Batch-condense 230 items into 11 condensed batches	Condensed knowledge
L3	ping_generator.py	Extract latest item per category + Z-axis value	ping_latest.json
L4	MoCKA-START.bat	Auto-execute ping_generator at session startup	DNA packet ready

2.3 Injection Mechanism

The generated packet is presented to the AI at session initialization as the first context element. The AI acknowledges receipt via ACK protocol:

```
[MOCKA]{...packet...} → AI responds: '応答（進化レバー適用）'
```

ACK confirms that the governance context has been loaded and behavioral constraints are active for the session duration.

3. Empirical Results

Metric	Pre-injection	Post-injection	Delta
Execution Rate	0.44	1.00	+0.56
Reproduction Rate	0.17	1.00	+0.83
INSTRUCTION_IGNORE incidents	4 per 10 sessions	0 per 10 sessions	−4
Session startup time overhead	0s	~2s	+2s

4. Limitations

Session-scope only: DNA injection controls behavior within a session. Cross-session persistence requires re-injection at each startup.

Model compliance: Injection relies on the model's instruction-following behavior. Models with low instruction adherence may not honor injected constraints.

Packet size: Larger packets consume context window capacity. Current implementation uses <500 bytes per packet.

5. Conclusion

The DNA Packet Injection protocol provides a practical solution to the session boundary problem in AI governance. By encoding institutional constraints as compact, auto-generated JSON packets and injecting them at session initialization, MoCKA achieves cross-session behavioral consistency without model fine-tuning. The four-layer pipeline (essence → condense → generate → inject) ensures packets reflect the latest operational knowledge at each session start.

References

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